

CHAPTER

10.3

EC-05

Duration : Three Hours
150

Maximum Marks :

Question 1- 30 Carry one Mark each.

1. The following differential equation has

$$3 \frac{d^2 y}{dt^2} + 4 \left(\frac{dy}{dt} \right)^3 + y^2 + 2 = x$$

- (A) degree = 2, order = 1
- (B) degree = 3, order = 2
- (C) degree = 4, order = 3
- (D) degree = 2, order = 3

2. Choose the function $f(t); -\infty < t < \infty$ for which a Fourier series cannot be defined.

- (A) $3 \sin(25t)$
- (B) $4 \cos(20t + 3) + 2 \sin(710t)$
- (C) $e^{-|t|} \sin(25t)$
- (D) 1

3. A fair dice is rolled twice. The probability that an odd number will follow on even number is

- (A) $\frac{1}{2}$
- (B) $\frac{1}{6}$
- (C) $\frac{1}{3}$
- (D) $\frac{1}{4}$

4. A solution of the following differential equation is given by

$$\frac{d^2 y}{dt^2} - 5 \frac{dy}{dt} + 6y = 0$$

- (A) $y = e^{2x} + e^{-3x}$
- (B) $y = e^{2x} + e^{3x}$
- (C) $y = e^{-2x} + e^{-3x}$
- (D) $y = e^{-2x} + e^{-3x}$

5. The function $x(t)$ is shown in the figure. Even and odd parts of a unit step function $u(t)$ are respectively,

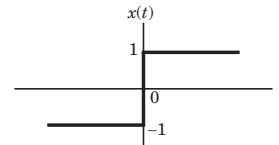


Fig. Q5

- (A) $\frac{1}{2}, \frac{1}{2} x(t)$
- (B) $-\frac{1}{2}, \frac{1}{2} x(t)$
- (C) $\frac{1}{2}, -\frac{1}{2} x(t)$
- (D) $-\frac{1}{2}, -\frac{1}{2} x(t)$

6. The region of convergence of z - transform of the sequence $\left(\frac{5}{6}\right)^n u(n) - \left(\frac{6}{5}\right)^n u(-n-1)$ must be

- (A) $|z| < \frac{5}{6}$
- (B) $|z| > \frac{5}{6}$
- (C) $\frac{5}{6} < |z| < \frac{6}{5}$
- (D) $\frac{6}{5} < |z| < \infty$

7. The condition on R, L and C such that the step response $y(t)$ in the figure has no oscillations, is

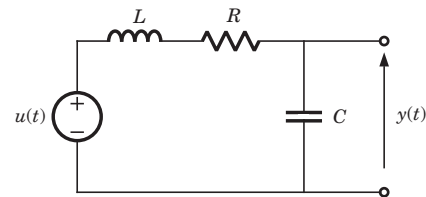


Fig. Q7

- (A) $R \geq \frac{1}{2} \sqrt{\frac{L}{C}}$
- (B) $R \geq \sqrt{\frac{L}{C}}$
- (C) $R \geq 2 \sqrt{\frac{L}{C}}$
- (D) $R = \sqrt{\frac{1}{LC}}$

8. The $ABCD$ parameters of an ideal $n:1$ transformer shown in the figure are $\begin{bmatrix} n & 0 \\ 0 & X \end{bmatrix}$. The value of x will be

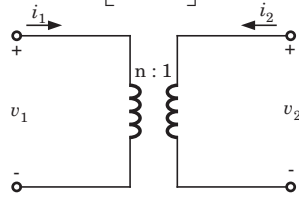


Fig. Q8

- (A) n (B) $\frac{1}{n}$
 (C) n^2 (D) $\frac{1}{n^2}$
9. In a series RLC circuit, $R=2 \text{ k}\Omega$, $L=1 \text{ H}$ and $C=\frac{1}{400} \mu$. The resonant frequency is
- (A) $2 \times 10^4 \text{ Hz}$ (B) $\frac{1}{\pi} \times 10^4 \text{ Hz}$
 (C) 10^4 Hz (D) $2\pi \times 10^4 \text{ Hz}$
10. The maximum power that can be transferred to the load resistor R_L from the voltage source in the figure is

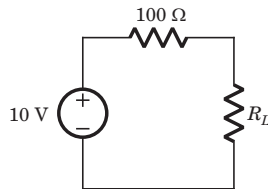


Fig. Q10

- (A) 1 W (B) 10 W
 (C) 0.25 W (D) 0.5 W
11. The bandgap of Silicon at room temperature is
- (A) 1.3 eV (B) 0.7 eV
 (C) 1.1 eV (D) 1.4 eV
12. A Silicon PN junction at a temperature of 20°C has a reverse saturation current of 10 pico - Amperes (pA). The reserve saturation current at 40°C for the same bias is approximately
- (A) 30 pA (B) 40 pA
 (C) 50 pA (D) 60 pA
13. The primary reason for the widespread use of Silicon in semiconductor device technology is

- (A) abundance of Silicon on the surface of the Earth.
 (B) larger bandgap of Silicon in comparison to Germanium.
 (C) favorable properties of Silicon - dioxide (SiO_2)
 (D) lower melting point.

14. The effect of current shunt feedback in an amplifier is to

- (A) increase the input resistance and decrease the output resistance.
 (B) increase both input and output resistance
 (C) decrease both input and output resistance.
 (D) decrease the input resistance and increase the output resistance.

15. The input resistance of the amplifier shown in the figure is

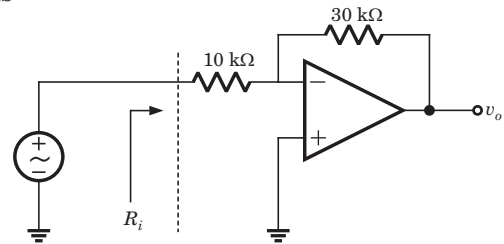


Fig. Q15

- (A) $\frac{30}{4} \text{ k}\Omega$ (B) 10 k Ω
 (C) 40 k Ω (D) infinite

16. The first and the last critical frequency of an RC - driving point impedance function must respectively be

- (A) a zero and a pole
 (B) a zero and a zero
 (C) a pole and a pole
 (D) a pole and a zero

17. The cascode amplifier is a multistage configuration of

- (A) CC - CB (B) CE - CB
 (C) CB - CC (D) CE - CC

18. Decimal 43 in Hexadecimal and BCD number system is respectively

- (A) B2, 0100 011
 (B) 2B, 0100 0011
 (C) 2B, 0011 0100
 (D) B2, 0100 0100

19. The Boolean function f implemented in the figure using two input multiplexes is

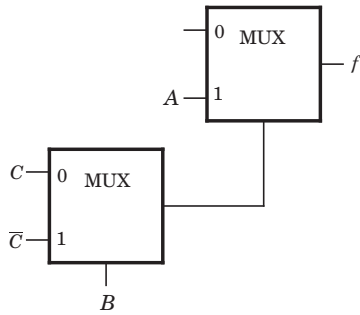
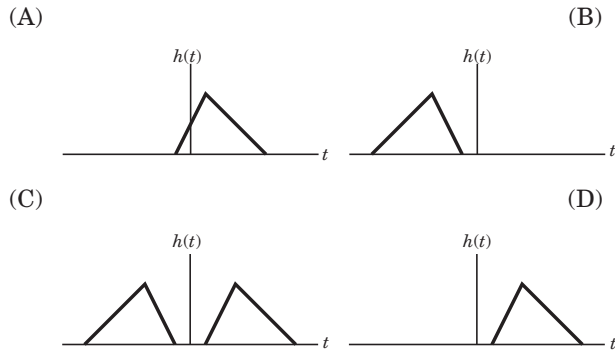


Fig. Q19

- (A) $\overline{A}BC + A\overline{B}\overline{C}$ (B) $ABC + \overline{A}\overline{B}\overline{C}$
 (C) $\overline{A}BC + \overline{A}\overline{B}\overline{C}$ (D) $\overline{A}BC + \overline{A}B\overline{C}$

20. Which of the following can be impulse response of a causal system?



21. Let $x(n) = \left(\frac{1}{2}\right)^n u(n)$, $y(n) = x^2(n)$ and $Y(e^{j\omega})$ be the Fourier transform of $y(n)$ then $Y(e^{j0})$ is

- (A) $\frac{1}{4}$ (B) 2
 (C) 4 (D) $\frac{4}{3}$

22. Find the correct match between group 1 and group 2

- | Group 1 | Group II |
|---------------------------------------|-------------------------|
| P. $\{1 + km(t)\} A \sin(\omega_c t)$ | W. Phase Modulation |
| Q. $km(t) A \sin(\omega_c t)$ | X. Frequency Modulation |
| R. $A \sin(\omega_c t + km(t))$ | Y. Amplitude Modulation |
| (A) P-Z, Q-Y, R-X, S-W | (B) P-W, Q-X, R-Y, S-Z |
| (C) P-X, Q-W, R-Z, S-Y | (D) P-Y, Q-Z, R-W, S-X |

23. The power in the signal

$$s(t) = 8 \cos\left(20\pi - \frac{\pi}{2}\right) + 4 \sin(15\pi t) \text{ is}$$

- (A) 40 (B) 41
 (C) 42 (D) 82

24. Which of the following analog modulation scheme requires the minimum transmitted power and minimum channel bandwidth?

- (A) VSB (B) DSB - SC
 (C) SSB (D) AM

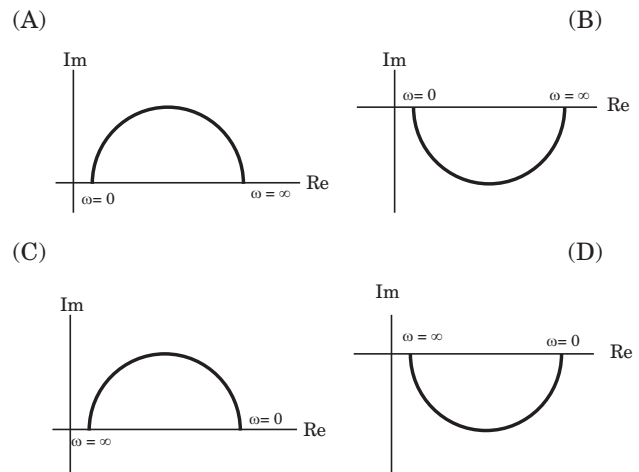
25. A linear system is equivalently represented by two sets of state equations:

$$\dot{X} = AX + BU \text{ And } \dot{W} = CW + DU$$

The eigenvalues of the representations are also computed as $[\lambda]$ and $[\mu]$. Which one of the following statements is true?

- (A) $[\lambda] = [\mu]$ and $X = W$ (B) $[\lambda] = [\mu]$ and $X \neq W$
 (C) $[\lambda] \neq [\mu]$ and $X = W$ (D) $[\lambda] = [\mu]$ and $X \neq W$

26. Which one of the following polar diagrams corresponds to a lag network?



27. Despite the presence of negative feedback, control systems still have problems of instability because the

- (A) Components used have non-linearities
 (B) Dynamic equations of the subsystem are not known exactly.
 (C) Mathematical analysis involves approximations.
 (D) System has large negative phase angle at high frequencies.

37. Given an orthogonal matrix

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix}, [AA^T]^{-1} \text{ is}$$

- (A) $\begin{bmatrix} \frac{1}{4} & 0 & 0 & 0 \\ 0 & \frac{1}{4} & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & \frac{1}{2} \end{bmatrix}$ (B) $\begin{bmatrix} \frac{1}{2} & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & \frac{1}{2} \end{bmatrix}$
- (C) $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ (D) $\begin{bmatrix} \frac{1}{4} & 0 & 0 & 0 \\ 0 & \frac{1}{4} & 0 & 0 \\ 0 & 0 & \frac{1}{4} & 0 \\ 0 & 0 & 0 & \frac{1}{4} \end{bmatrix}$

38. For the circuit shown in the figure, the instantaneous current $i_i(t)$ is

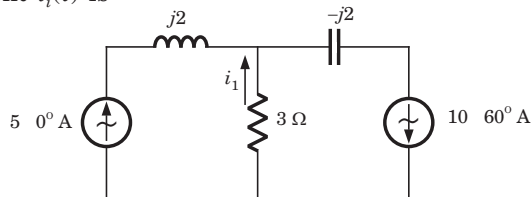


Fig. Q38

- (A) $\frac{10\sqrt{3}}{2} \angle 90^\circ$ Amps. (B) $\frac{10\sqrt{3}}{2} \angle -90^\circ$ Amps.
- (C) $5 \angle 60^\circ$ Amps (D) $5 \angle -60^\circ$ Amps

39. Impedance Z as shown in the given figure is

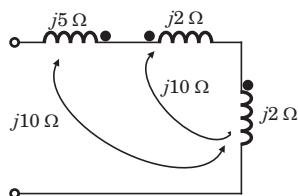


Fig. Q39

- (A) $j29 \Omega$ (B) $j9 \Omega$
- (C) $j19 \Omega$ (D) $j39 \Omega$

40. For the circuit shown in figure, Thevenin's voltage and Thevenin's equivalent resistance at terminals a - b is

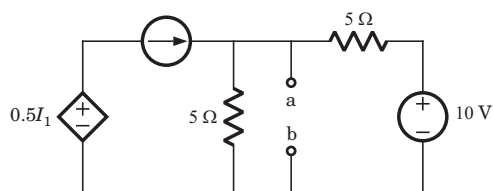


Fig. Q40

- (A) 5 V and 2Ω (B) 7.5 V and 2.5Ω
- (C) 4 V and 2Ω (D) 3 V and 2.5Ω

41. If $R_1 = R_2 = R_3 = R$ and $R_4 = 1.1R$ in the bridge circuit shown in the figure, then the reading in the ideal voltmeter connected between a and b is

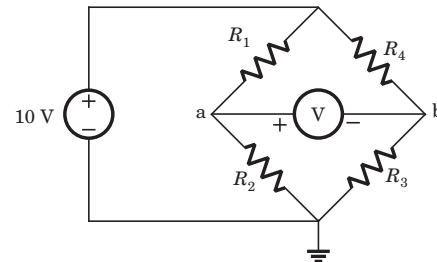


Fig. Q41

- (A) 0.238 V (B) 0.138 V
- (C) -0.238 V (D) 1 V.

42. The h parameters of the circuit shown in the figure are

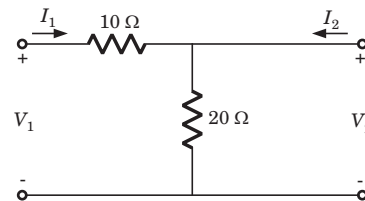


Fig. Q42

- (A) $\begin{bmatrix} 0.1 & 0.1 \\ -0.1 & 0.3 \end{bmatrix}$ (B) $\begin{bmatrix} 10 & -1 \\ 1 & 0.05 \end{bmatrix}$
- (C) $\begin{bmatrix} 30 & 20 \\ 20 & 20 \end{bmatrix}$ (D) $\begin{bmatrix} 10 & 1 \\ -1 & 0.05 \end{bmatrix}$

43. A square pulse of 3 volts amplitude is applied to C-R circuit shown in the figure. The capacitor is initially uncharged. The output voltage V_o at time $t = 2$ sec is

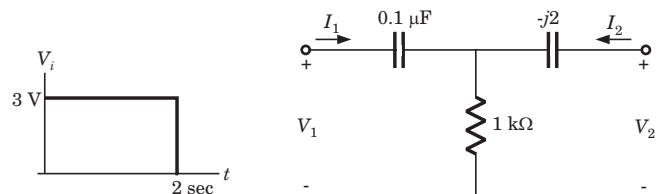


Fig. Q43

- (A) 3 V (B) -3 V
- (C) 4 V (D) -4 V

44. A Silicon sample A is doped with 10^{18} atoms/cm³ of boron. Another sample b of identical dimension is doped with 10^{18} atoms/cm³ phosphorus. The ratio of electron to

hole mobility is 3. The ratio of conductivity of the sample A to B is

- (A) 3 (B) $\frac{1}{3}$
(C) $\frac{2}{3}$ (D) $\frac{3}{2}$

45. A Silicon PN junction diode under reverse bias has depletion region of width $10 \mu\text{m}$. The relative permittivity of Silicon, $\epsilon_r = 11.7$ and the permittivity of free space $\epsilon_o = 8.85 \times 10^{-12} \text{ F/m}$. The depletion capacitance of the diode per square meter is

- (A) $100 \mu\text{F}$ (B) $10 \mu\text{F}$
(C) $1 \mu\text{F}$ (D) $20 \mu\text{F}$

46. For an npn transistor connected as shown in figure $V_{BE} = 0.7 \text{ volts}$. Given that reverse saturation current of the junction at room temperature 300 K is 10^{-13} A , the emitter current is

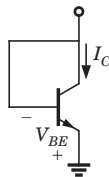


Fig. Q46

- (A) 30 mA (B) 39 mA
(C) 49 mA (D) 20 mA

47. The voltage e_o is indicated in the figure has been measured by an ideal voltmeter. Which of the following can be calculated ?

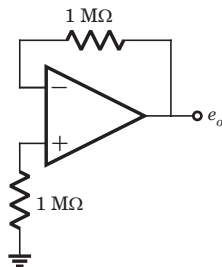


Fig. Q47

- (A) Bias current of the inverting input only
(B) Bias current of the inverting and non-inverting inputs only
(C) Input offset current only
(D) Both the bias currents and the input offset current.

48. The OP-amp circuit shown in the figure is filter. The type of filter and its cut. Off frequency are respectively.

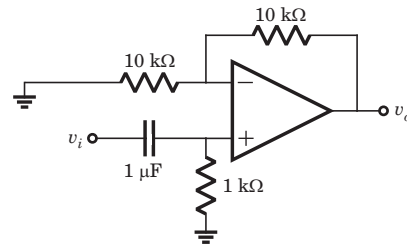


Fig. Q48

- (A) high pass, 1000 rad/sec .
(B) Low pass, 1000 rad/sec .
(C) high pass, 1000 rad/sec .
(D) low pass, 10000 rad/sec .

49. In an ideal differential amplifier shown in the figure, a large value of (R_E)

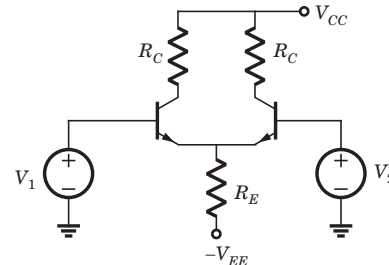


Fig. Q49

- (A) increase both the differential and common - mode gains
(B) increases the common mode gain only
(C) decreases the differential mode gain only
(D) decreases the common mode gain only.

50. For an n-channel MOSFET and its transfer curve shown in the figure, the threshold voltage is

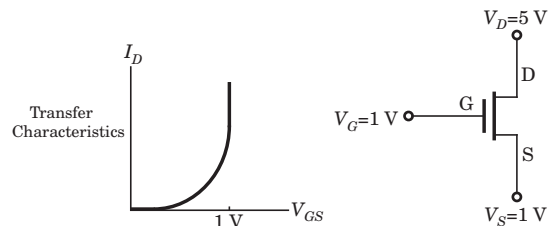


Fig. Q50

- (A) 1 V and the device is in active region
(B) -1 V and the device is in saturation region
(C) 1 V and the device is in saturation region
(D) -1 V and the device is in active region.

57. The given figure shows a ripple counter using positive edge triggered flip-flops. If the present state of the counter is $Q_2Q_1Q_0 = 001$ then its next state $Q_2Q_1Q_0$ will be

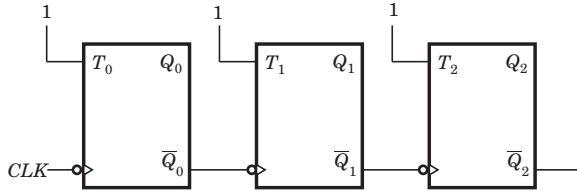


Fig. Q57

- (A) 010 (D) 100
(B) 111 (D) 101

58. What memory address range is NOT represents by chip # 1 and chip # 2 in the figure A_0 to A_{15} in this figure are the address lines and CS means chip select.

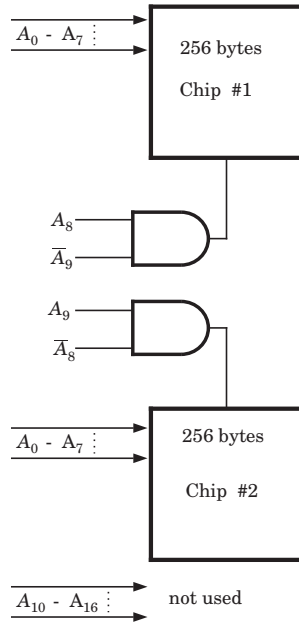


Fig. Q58

- (A) 0100 - 02FF. (B) 1500 - 16FF
(C) F900-FAFF (D) F800 - F9FF

59. The output $y(t)$ of a linear time invariant system is related to its input $x(t)$ by the following equation

$$y(t) = 0.5x(t - t_d + T) + x(t - t_d) + 0.5x(t - t_d + T)$$

The filter transfer function $H(\omega)$ of such a system is given by

- (A) $(1 + \cos \omega T)e^{-j\omega t_d}$ (B) $(1 + 0.5 \cos \omega T)e^{-j\omega t_d}$
(C) $(1 - \cos \omega T)e^{-j\omega t_d}$ (D) $(1 - 0.5 \cos \omega T)e^{-j\omega t_d}$

60. Match the following and choose the correct combination.

Group 1

- E. Continuous and periodic signal
F. Continuous and periodic signal
G. Discrete and aperiodic signal
H. Discrete and periodic signal

Group 2

- Fourier representation is continuous and aperiodic
 - Fourier representation is discrete and aperiodic
 - Fourier representation is continuous
 - Fourier representation is discrete and periodic
- (A) E-3, F-2, G-4, H-1 (A) E-1, F-3, G-2, H-4
(C) E-1, F-2, G-3, H-4 (D) E-2, F-1, G-4, H-3

61. A signal $x(n) = \sin(\omega_0 n + \phi)$ is the input to a linear time-invariant system having a frequency response $H(e^{j\omega})$. If the output of the system is $Ax(n - n_0)$ then the most general form of will be

- (A) $-n_0\omega_0 + \beta$ for any arbitrary real
(B) $-n_0\omega_0 + 2\pi k$ for any arbitrary integer k
(C) $n_0\omega_0 + 2\pi k$ for any arbitrary integer k
(D) $-n_0\omega_0\phi$

62. For a signal the Fourier transform is $X(f)$. Then the inverse Fourier transform of $X(3f + 2)$ is given by

- (A) $\frac{1}{2}x\left(\frac{t}{2}\right)e^{j3\pi t}$ (B) $\frac{1}{3}x\left(\frac{t}{3}\right)e^{-\frac{j4\pi t}{3}}$
(C) $3x(3t)e^{-j4\pi t}$ (D) $x(3t + 2)$

63. The polar diagram of a conditionally stable system for open loop gain $K = 1$ is shown in the figure. The open loop transfer function of the system is known to be stable. The closed loop system is stable for

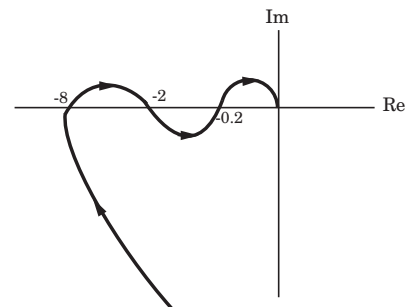


Fig. Q63

- (A) $K < 5$ and $\frac{1}{2} < K < \frac{1}{8}$ (B) $K < \frac{1}{8}$ and $\frac{1}{2} < K < 5$
 (C) $K < \frac{1}{8}$ and $5 < K$ (D) $K > \frac{1}{8}$ and $5 > K$

64. In the derivation of expression for peak percent overshoot

$$M_p = \exp\left(\frac{-\pi\xi}{\sqrt{1-\xi^2}}\right) \times 100\%$$

Which one of the following conditions is NOT required?

- (A) System is linear and time invariant
 (B) The system transfer function has a pair of complex conjugate poles and no zeroes.
 (C) There is no transportation delay in the system.
 (D) The system has zero initial conditions.

65. Given the ideal operational amplifier circuit shown in the figure indicate the correct transfer characteristics assuming ideal diodes with zero cut-in voltage.

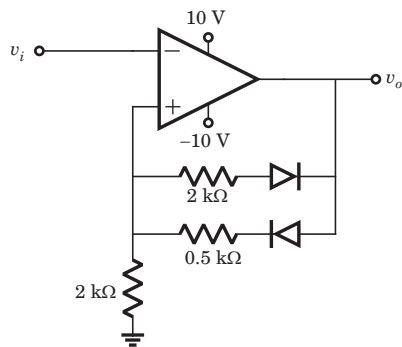
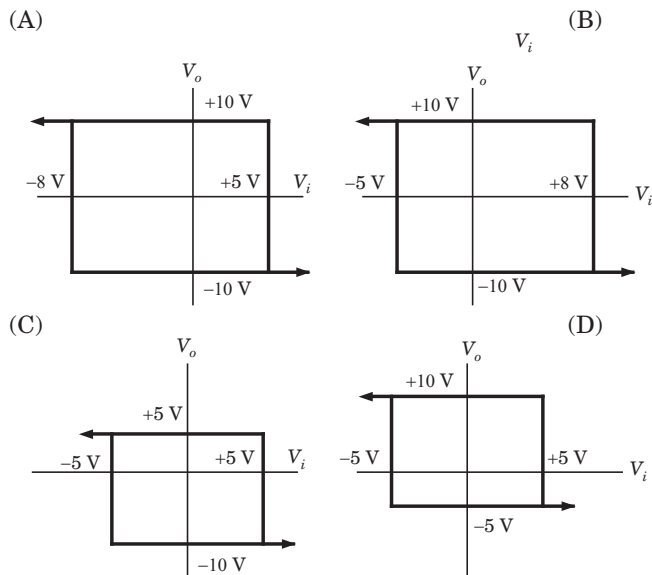


Fig. Q65



- 66.** A ramp input applied to an unity feedback system results in 5% steady state error. The type number and zero frequency gain of the system are respectively
 (A) 1 and 20 (B) 0 and 20
 (C) 0 and $\frac{1}{20}$ (D) 1 and $\frac{1}{20}$

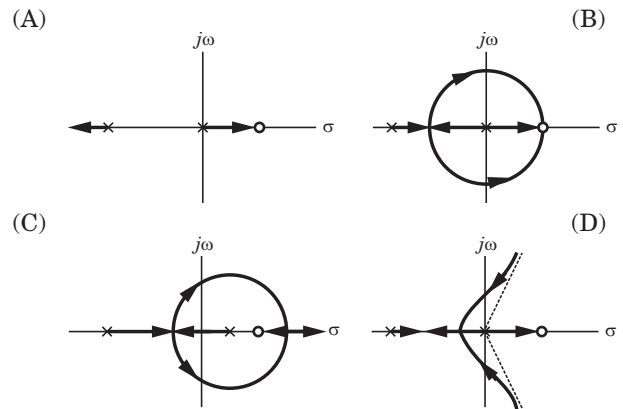
67. A double integrator plant $G(s) = K/s^2$, $H(s) = 1$ is to be compensated to achieve the damping ratio and undamped natural frequency, $\omega = 5$ rad/s which one of the following compensator $G_c(s)$ will be suitable?

- (A) $\frac{s+3}{s+9.9}$ (B) $\frac{s+9.9}{s+3}$
 (C) $\frac{s-6}{s+8.33}$ (D) $\frac{s-6}{s}$

68. An unity feedback system is given as

$$G(s) = \frac{K(1-s)}{s(s+3)}$$

Indicate the correct root locus diagram.



69. A MOS capacitor made using P type substrate is in the accumulation mode. The dominant charge in the channel is due to the presence of

- (A) holes
 (B) electrons
 (C) positively charged ions
 (D) negatively charged ions

70. A device with input $x(t)$ and output $y(t)$ is characterized $y(t) = x^2(t)$. An FM signal with frequency deviation of 90 kHz and modulating signal bandwidth of 5 kHz is applied to this device. The bandwidth of the output signal is

- (A) 370 kHz (B) 190 kHz
 (C) 380 kHz (D) 95 kHz

COMMON DATA QUESTION 78, 79, 80:

Given, $r_d = 20 \text{ k}\Omega$, $I_{DSS} = 10 \text{ mA}$, $V_p = -8 \text{ V}$

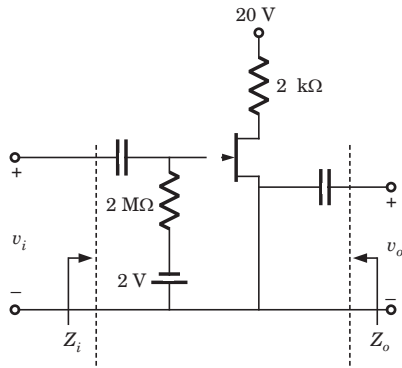


Fig. Q78

- 78.** Z_i and Z_o of the circuit are respectively
 (A) $2 \text{ M}\Omega$ and $2 \text{ k}\Omega$ (B) $2 \text{ M}\Omega$ and $\frac{20}{11} \text{ k}\Omega$
 (C) ∞ and $2 \text{ k}\Omega$ (D) ∞ and $\frac{20}{11} \text{ k}\Omega$
- 79.** I_D and V_{DS} under DC conditions are respectively
 (A) 5.625 mA and 8.75 V (B) 7.500 mA and 5.00 V
 (C) 4.500 mA and 11.00 V (D) 6.250 mA and 7.50 V
- 80.** Transconductance in milli-Siemens (mS) and voltage gain of the amplifier are respectively
 (A) 1.875 mS and 3.41 (B) 1.875 mS and -3.41
 (C) 3.3 mS and -6 (D) 3.3 mS and 6

Linked Answer Questions : Q.81a to 85b Carry Two Marks Each

Statement For Linked Answer Questions 81a and 81b:

Consider an 8085 microprocessor system.

81a. The following program starts at location 0100H.
 LXI SP, OOFF
 LXI H, 0701
 MVI A, 20H
 SUB M

The content of accumulator when the program counter reaches 0109 H is

- (A) 20 H (B) 02 H
 (C) 00 H (D) FF H

81b. If in addition following code exists from 019H onwards,
 ORI 40 H
 ADD M

What will be the result in the accumulator after the last instruction is executed?

- (A) 40 H (B) 20 H
 (C) 60 H (D) 42 H

Statement for Linked Answer Question 82a and 82b:

The dopen loop transfer function of a unity feedback system is given by

82a. The gain and phase crossover frequencies in rad/sec are, respectively
 (A) 0.632 and 1.26 (B) 0.632 and 0.485
 (C) 0.485 and 0.632 (D) 1.26 and 0.632

82b. Based on the above results, the gain and phase margins of the system will be
 (A) -7.09 dB and 87.5° (B) 7.09 dB and 87.5°
 (C) 7.09 dB and -87.5° (D) -7.09 and -87.5°

Statement for linked answer question 83a and 83b

Asymmetric three - level midtread quantizer is to be designed assuming equiprobable occurrence of all quantization levels.

83a. If the probability density function is divided into three regions as shown in the figure, the value of a in the figure is

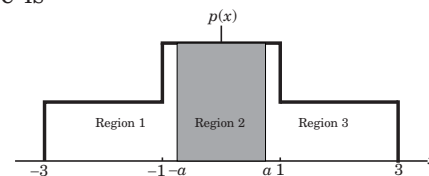


Fig. Q83

- (A) $\frac{1}{3}$ (B) $\frac{2}{3}$
 (C) $\frac{1}{2}$ (D) $\frac{1}{4}$

83b. The quantization noise power for the quantization region between $-a$ and $+a$ in the figure is

- (A) $\frac{4}{81}$ (B) $\frac{1}{9}$
 (C) $\frac{5}{81}$ (D) $\frac{2}{81}$

Statement of Linked Answer Questions 84a and 84b

Voltage standing wave pattern in a lossless transmission line with characteristic impedance 50 and a resistive load is shown in the figure.

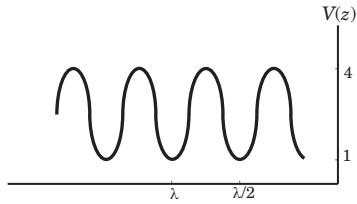


Fig. Q84a and Q84b

84a. The value of the load resistance is

- (A) 50 Ω (B) 200 Ω
(C) 12.5 Ω (D) 0

84b. The reflection coefficient is given by

- (A) -0.6 (B) -1
(C) 0.6 (D) 0

Statement of Linked Answer Question 85a and 85b:

A sequence $x(n)$ has non-zero values as shown in the figure.(A)

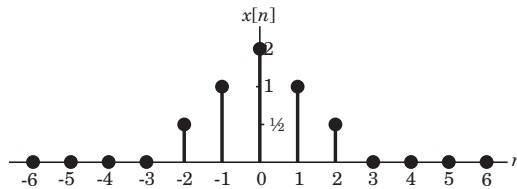
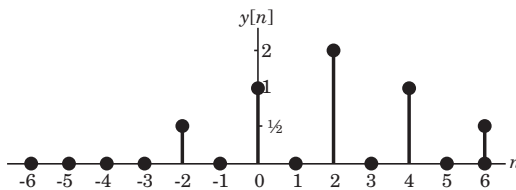


Fig. Q85

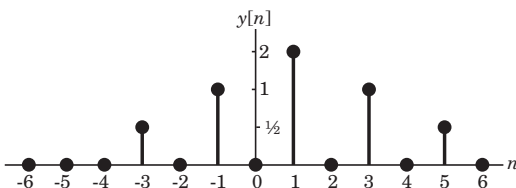
85a. The sequence

$$y[n] = \begin{cases} x\left(\frac{n}{2} - 1\right) & \text{for } n \text{ even} \\ 0 & \text{for } n \text{ odd} \end{cases} \quad \text{will be}$$

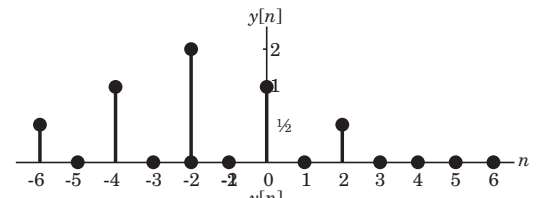
(A)



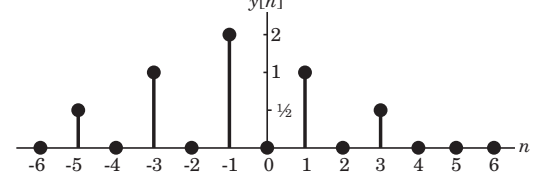
(B)



(C)



(D)



85b. The Fourier transform of $y[2n]$ will be

- (A) $e^{-2j\omega} [\cos 4\omega + 2 \cos 2\omega + 2]$
(B) $[\cos 2\omega + 2 \cos \omega + 2]$
(C) $e^{-j\omega} [\cos 2\omega + 2 \cos \omega + 2]$
(D) $e^{-2j\omega} [\cos 2\omega + 2 \cos \omega + 2]$
